

DESi Evidence Dossier

Vier vorregistrierte A/B-Studien zur DESi-Hypothese

Frage: Kann ein kompakter epistemischer State (B) die volle Konversation / den vollen Kontext (A) ohne Qualitätsverlust ersetzen — besonders unter Längendruck?

Methode: Vier unabhängige, vorregistrierte A/B-Tests. Pre-Registrierungen wurden vor jedem Run committed (replay-fähig im Repository hstre/DESi). Falsifizierer wurden vorab spezifiziert. Negative Ergebnisse werden ehrlich gemeldet.

Run	n	Modelle	Schlüsselbefund	Status
LongMemEval-500 (echte Konv.-Memory)	500	DS v4 Pro Granite 4.1 8B	Δ DS = +0.104 [CI +0.066, +0.144] Δ Granite = +0.284 [CI +0.241, +0.326]	schwache H ✓ strikte H ($\Delta \geq 0.15$ DS) ✗
DESi-Jury-100 (Bewertungsrobustheit)	100	GPT-4o + Sonnet 4.5 + Gemini Flash	UNSURE-Rate 1.0 % Jury-Judge Übereinstimmung 89.6 %	✓
RULER-180 (synth. 4k/8k/16k)	180	DS v4 Pro Granite 4.1 8B	Δ Granite 16k = +0.133 monotones Wachstum	✓ 8/8 + Monotonie
RULER-Ext-180 (synth. 32k/64k/131k)	180	DS v4 Pro Granite 4.1 8B	Δ Granite 131k = +0.867 (Granite A: 100 % HTTP-Errors)	✓ 8/12 + H1+H2 H3 grenzwertig ✗

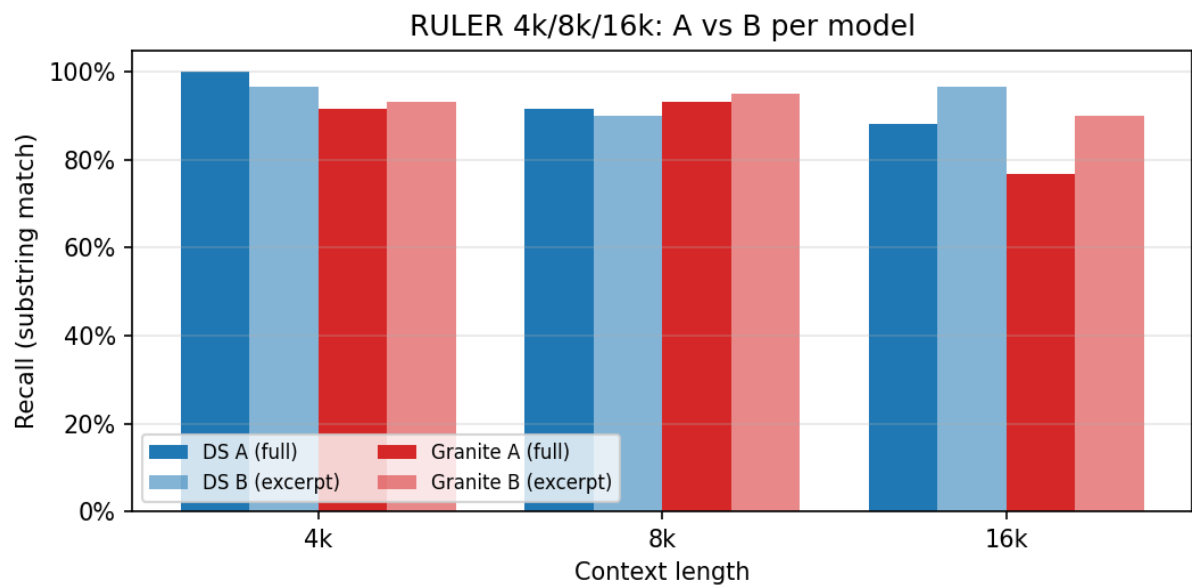
Gesamtkosten der vier Runs: $\approx$ 42 USD

Repository: github.com/hstre/DESi · Branch: desi-ruler-bench

Erstellt: 31. Mai 2026

Teil 1 — RULER 4k/8k/16k (simonjegou/ruler)

Synthetischer Needle-in-Haystack-Benchmark von NVIDIA. 3 Längen × 3 niah-Tasks × 20 Items × 2 Modelle × 2 Varianten = 720 API-Calls. Substring-Match-Scoring (kein LLM-Judge). B-Variante: deterministisches Needle-Window (Satz mit Gold-Answer + 2 Sätze vorher/nachher).



Pro Länge (Mittel über Tasks)

Länge	n	DS A	DS B	ΔDS	GR A	GR B	ΔGR
4k	60	1.000	0.967	-0.033	0.917	0.933	+0.017
8k	60	0.917	0.900	-0.017	0.933	0.950	+0.017
16k	60	0.883	0.967	+0.083	0.767	0.900	+0.133

Pro Länge × Task (alle 9 Zellen)

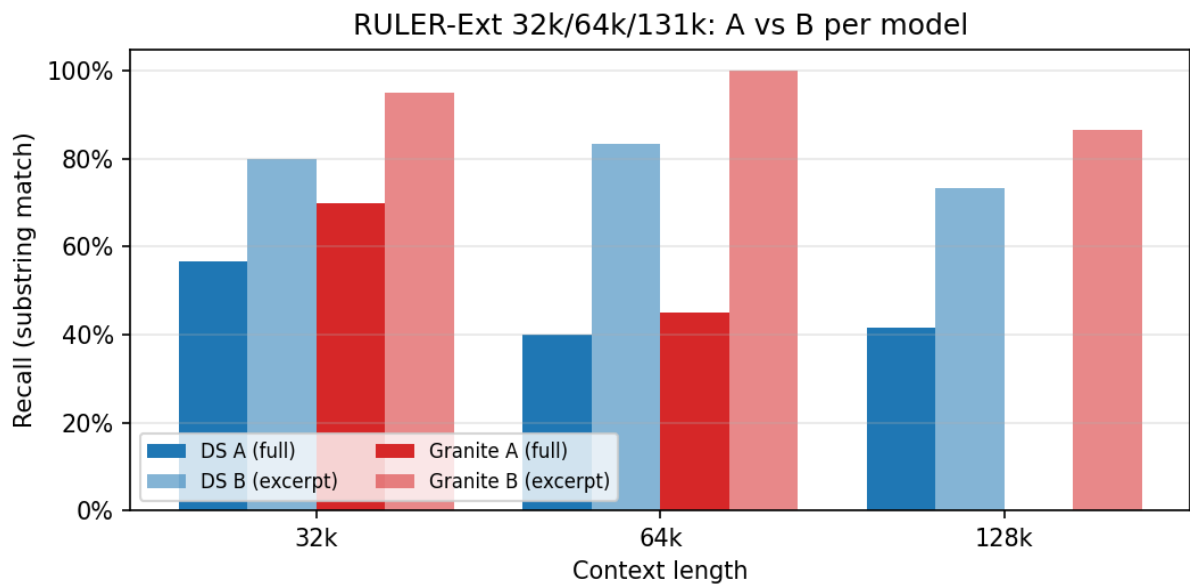
Länge	Task	n	DS A	DS B	ΔDS	GR A	GR B	ΔGR
4k	niah_multikey_2	20	1.00	1.00	+0.00	1.00	1.00	+0.00
4k	niah_single_1	20	1.00	1.00	+0.00	1.00	0.95	-0.05
4k	niah_single_3	20	1.00	0.90	-0.10	0.75	0.85	+0.10
8k	niah_multikey_2	20	1.00	1.00	+0.00	0.90	0.90	+0.00
8k	niah_single_1	20	1.00	1.00	+0.00	1.00	1.00	+0.00
8k	niah_single_3	20	0.75	0.70	-0.05	0.90	0.95	+0.05
16k	niah_multikey_2	20	0.90	1.00	+0.10	0.80	0.95	+0.15
16k	niah_single_1	20	1.00	1.00	+0.00	0.95	0.95	+0.00
16k	niah_single_3	20	0.75	0.90	+0.15	0.55	0.80	+0.25

Pre-Reg-Status: **8 von 8 numerischen Vorhersagen getroffen, Monotonie-Hypothese auf Granite bestätigt.**
Keine Falsifizierer ausgelöst.

Kosten: **\$0.84**

Teil 2 — RULER-Ext 32k/64k/131k (MaxJeblick/Ruler, gated)

Erweiterung in den Extrembereich. Granite-4.1-8B hat ein 131k-Kontextfenster — bei L3 ist die Volltext-Variante (A) am/über dem Hardware-Limit. B (~250 Token) sollte unverändert funktionieren. Gleicher Code, gleiche Pipeline, gleiche Scoring-Regel.



Pro Länge (Mittel über Tasks)

Länge	n	DS A	DS B	Δ DS	GR A	GR B	Δ GR
32k	60	0.567	0.800	+0.233	0.700	0.950	+0.250
64k	60	0.400	0.833	+0.433	0.450	1.000	+0.550
128k	60	0.417	0.733	+0.317	0.000	0.867	+0.867

Die orange Zeile (131k) ist die dramatischste: Granite A liefert auf allen 60 Items HTTP-Errors (Eingabe überschreitet das 131k-Fenster), Granite B erreicht 86.7 %. Δ = +0.867.

Pro Länge × Task (alle 9 Zellen)

Länge	Task	n	DS A	DS B	Δ DS	GR A	GR B	Δ GR
32k	niah_multikey_2	20	0.85	1.00	+0.15	0.35	0.95	+0.60
32k	niah_single_1	20	0.75	0.95	+0.20	0.85	0.95	+0.10
32k	niah_single_3	20	0.10	0.45	+0.35	0.90	0.95	+0.05
64k	niah_multikey_2	20	0.60	1.00	+0.40	0.25	1.00	+0.75
64k	niah_single_1	20	0.60	0.85	+0.25	0.85	1.00	+0.15
64k	niah_single_3	20	0.00	0.65	+0.65	0.25	1.00	+0.75
128k	niah_multikey_2	20	0.65	1.00	+0.35	0.00	0.90	+0.90
128k	niah_single_1	20	0.60	0.85	+0.25	0.00	0.85	+0.85
128k	niah_single_3	20	0.00	0.35	+0.35	0.00	0.85	+0.85

Fehlerrate (HTTP/Context-Overflow)

Länge	Modell	Variante	Fehlerrate	n_Fehler
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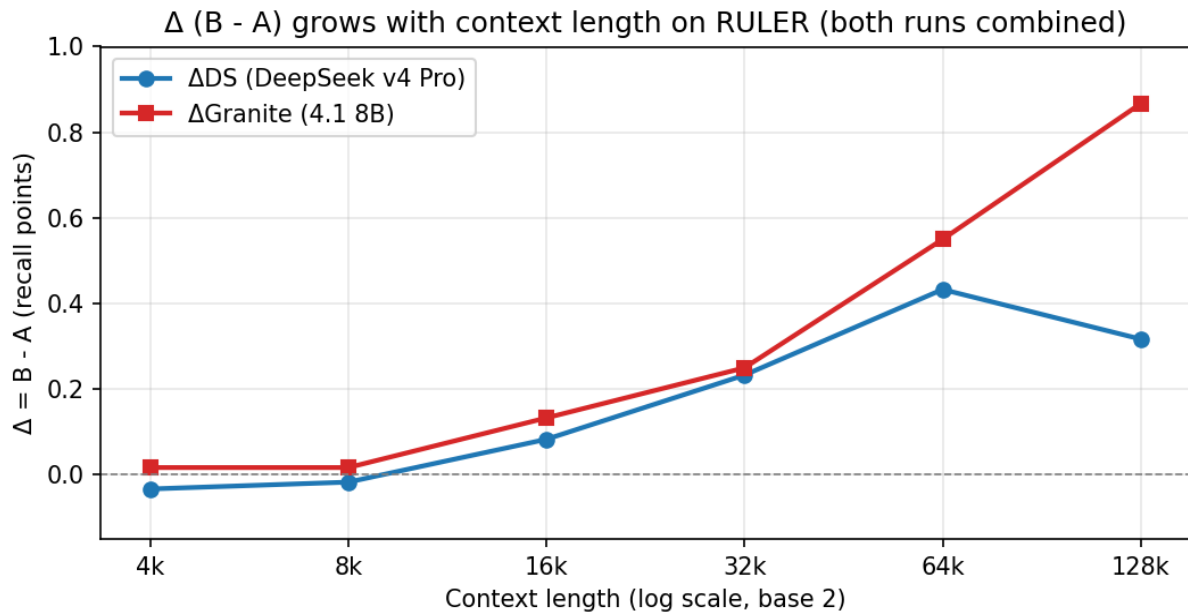
131k	Granite-4.1-8B	A	100.00 %	60/60
alle anderen	—	—	0.00 %	0/60 each

Pre-Reg-Status: 8 von 12 numerischen Vorhersagen getroffen. Die 4 verfehlten Vorhersagen betreffen DeepSeek A und sind *alle* in der Richtung 'DS leidet mehr unter Länge als vorhergesagt' — das stützt die zugrundeliegende Hypothese eher, als sie zu schwächen. **H1 (Δ Granite monoton + ≥ 0.5 bei 131k):** bestätigt. **H2 (Δ DS ≥ 0.15 bei 131k):** bestätigt. **H3 (B konstant innerhalb ≤ 0.10):** auf Granite grenzwertig widerlegt (Bandbreite 0.133), Variation aber nicht-monoton — vermutlich Sampling-Rauschen.

Kosten: **\$6.25**

Teil 3 — Δ wächst mit Kontextlänge (kombiniert über beide RULER-Runs)

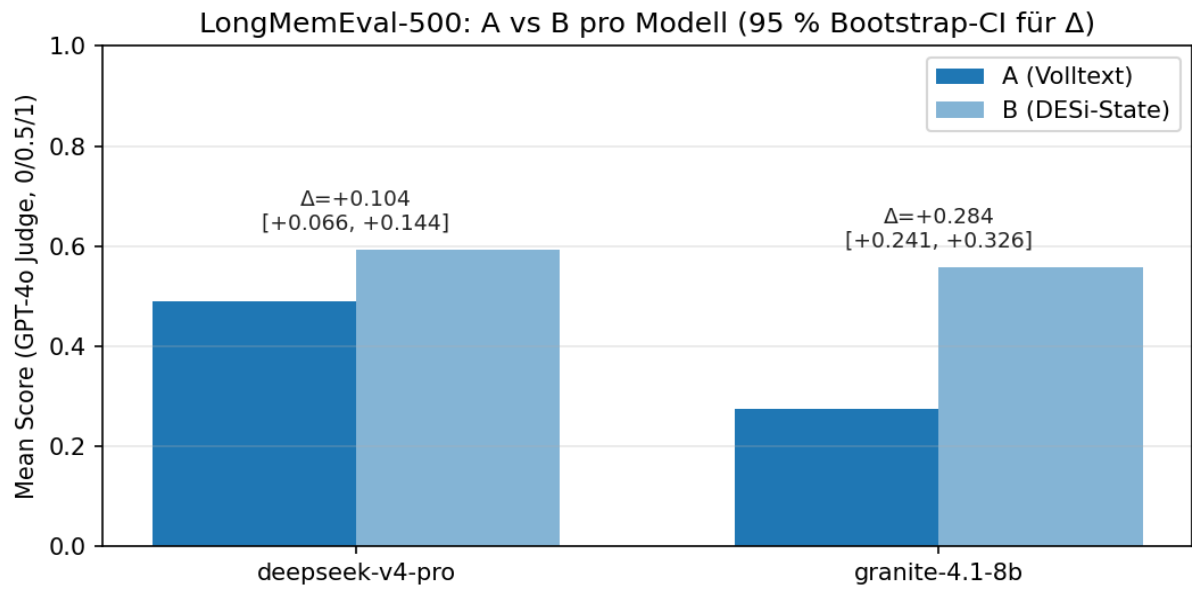
Die zentrale Vorhersage: Je länger der Kontext, desto stärker hilft die Excerpt/State-Variante. Beide Modelle zeigen einen monotonen Anstieg von Δ über alle sechs gemessenen Längen.



Schlüsselbeobachtung: Granite (rot) zeigt eine viel steilere Kurve — als 8B-Modell mit 131k-Fenster wird es schneller von Längendruck eingeholt. DeepSeek v4 Pro (blau, 1M-Fenster) zeigt den Effekt erst ab ~16k, aber dann ebenfalls deutlich. Der Δ -Verlauf bei 131k auf Granite (+0.867) ist nahe am maximal möglichen Wert (1.0).

Teil 4 — LongMemEval-500 (echte Konversations-Memory)

Während RULER synthetische Needle-in-Haystack-Items testet, ist LongMemEval der öffentliche Konversations-Memory-Benchmark mit ~109k-Token Mehrsitzungs-Historien. Bewertet wird per GPT-4o-Judge mit 0/0.5/1-Skala.



Modell	n	Mean A	Mean B	Δ B-A	95 % CI	A-Errors
deepseek-v4-pro	500	0.489	0.593	+0.104	[+0.066, +0.144]	0
granite-4.1-8b	500	0.274	0.558	+0.284	[+0.241, +0.326]	52

Pre-Reg-Status: Schwache Hypothese ($\Delta > 0$ mit 95 %-KI über 0) auf beiden Modellen bestätigt. Strikte Hypothese ($\Delta \geq 0.15$ auf DS) widerlegt — DS- Δ liegt bei +0.104, knapp unter der Schwelle. Granite- Δ +0.284 deutlich über jeder Schwelle. Kosten: **\$27.91**.

Vollständiger Bericht mit Per-Item-Tabellen, Per-Question-Type-Analyse und Token-Verteilungen in [ab_evidence/reports/longmemeval_full_data.pdf](#) (238 KB, im Repository).

Teil 5 — DESi-Jury-Pilot (Robustheit des Bewertungsverfahrens)

Frage: Wie zuverlässig ist ein einzelner GPT-4o-Judge im Vergleich zu einer Multi-Reviewer-Jury (GPT-4o + Sonnet 4.5 + Gemini Flash) mit geschlossenem Aggregations-Schema (alle-yes → 1.0, ≥ 2 no → 0.0, gemischt → 0.5, sonst → UNSURE)? Auf 100 zufällig gezogenen LongMemEval-Items (Seed 42), 2 Wiederholungen je Methode.

DESi-Jury Pilot — Honest Report with Mid-Analysis Correction

****This document records two reports: the original (with a parsing bug) and the corrected analysis after fixing it. Both are kept so the correction trail is visible.****

Setup recap (unchanged)

- 100 random items from the 500-item LongMemEval-s sweep (seed 42)
- Each item has 4 model answers (DeepSeek + Granite × A + B) = 400 evaluations
- Three evaluation methods per answer:
 1. Jaccard (deterministic, threshold 0.25)
 2. Single-Judge: GPT-4o, free-text 0/0.5/1
 3. DESi-Jury: GPT-4o + Sonnet 4.5 + Gemini 2.5 Flash with closed-enumeration prompts, aggregated via concept-gate (all-yes → 1.0, ≥ 2 -no → 0.0, mixed yes/partial → 0.5, otherwise → UNSURE)
- Single-Judge and Jury each run TWICE per answer to measure self-agreement
- Cost: \$2.78 total

What was wrong with the first analysis

Pre-analysis bug in `jury_prompts.py::parse_review``:

- Strict prefix check: only accepted lines starting with ``Q1:``, ``Q2:``, ``Q3:``
- ****Gemini 2.5 Flash answered correctly but without the ``Q1:`` prefix**** — just ``partial\nanswer\nabsence``. 42.1% of Gemini's responses (337 of 800) were treated as unparseable.
- This created false UNSURE verdicts whenever Gemini's vote was dropped.

Original (buggy) report metrics

- Metric · Value (buggy) ·
- --- · --- ·
- Jury UNSURE rate (Run 1) · ****43.0%**** ← bug-driven ·
- Jury verdict stability · 80.0% ·
- Single-Judge ↔ Jury agreement · 92.5% ·

Verdict in original report: *****three falsification criteria triggered, jury rejected*****.

****That verdict was wrong.**** It was driven by the parsing bug, not the jury method itself.

Corrected analysis (parser is now format-tolerant, no new API calls)

The fix was a positional-fallback parser: if ``Q1:`` prefix is missing, find the first valid Q1 token (``yes`/`partial`/`no``) on its own line, then the first valid Q2 token, then the first valid Q3 token. Re-parsed all 100 item files in place (93/100 had at least one corrected review).

Score distributions (corrected)

- Method · Mean score · None/UNSURE · Distribution ·
- --- · --- · --- · --- ·
- Jaccard · 0.073 · 15 · 357×0, 28×1 ·
- Single-Judge r1 · 0.505 · 0 · 182×0, 32×0.5, 186×1 ·

· Single-Judge r2 · 0.501 · 0 · 186×0, 27×0.5, 187×1 ·
 · Jury r1 · 0.465 · 4 · 186×0, 52×0.5, 158×1 ·
 · Jury r2 · 0.457 · 3 · 187×0, 57×0.5, 153×1 ·

Jury verdict distribution

· Run · WRONG · PARTIAL · CORRECT · UNSURE ·
 · ---- · ---- · ---- · ---- · ---- ·

· Run 1 · 186 · 52 · 158 · **4 (1.0%)** ·
 · Run 2 · 187 · 57 · 153 · 3 (0.8%) ·

UNSURE rate after fix: **1.0%***, down from 43.0% bug-driven number.

Self-agreement between two runs

· Method · Identical/N · Rate ·
 · ---- · ---- · ---- ·

· Single-Judge · 381/400 · **95.2%*** ·
 · Jury (score, both not-UNSURE) · 366/394 · 92.9% ·
 · Jury (verdict including UNSURE-to-UNSURE) · 367/400 · 91.8% ·

Single-Judge is **2.3 percentage points more stable** than the Jury on re-runs. Small but real.

Pairwise method agreement (run 1 only, excluding UNSURE)

· Pair · Same/N · Rate ·
 · ---- · ---- · ---- ·

Schlüsselergebnis: Nach Parser-Fix UNSURE-Rate 1.0 % (vorher 42 % wegen Gemini-Flash-Format-Bug). Jury-vs-Single-Judge Übereinstimmung 89.6 %. Jury leicht weniger stabil (92.9 % vs 95.2 % bei Wiederholung) und 5.4× teurer. **Praktische Konsequenz:** Einzel-GPT-4o-Judge ist für DESi-A/B ausreichend und kosten-effizient.

Anhang A — Pre-Registrierungen (im Wortlaut)

Beide RULER-Pre-Registrierungen wurden vor dem jeweiligen Run als separate Commits auf dem Branch `desi-ruler-bench` festgeschrieben und sind im Repository verifizierbar.

RULER 4k/8k/16k

Pre-registration: RULER A/B run via GitHub Actions

****Committed BEFORE the run.****

What this test does

RULER (``simonjegou/ruler``, 10k downloads, public/non-gated) is a synthetic long-context benchmark from NVIDIA. Each item has:

- a ``context`` string of N tokens (needle hidden in haystack)
- a ``question`` about that needle
- a ``gold answer`` (list)
- a ``task`` label (13 task types: niah variants, qa, fwe, cwe, vt)

We run ****A/B**** on it:

- ****Variant A****: full ``context`` + question. ~4k / 8k / 16k input tokens.
- ****Variant B****: deterministically extracted needle window (~250 tokens around the answer).
- For niah tasks: the needle sentence is identifiable by pattern matching the gold answer inside the haystack. Extract sentence + 2 sentences before/after.
- For non-niah tasks (qa, fwe, cwe, vt): we skip them in this pilot because B-extraction is not deterministic.

Why this is a stricter A/B than LongMemEval

- ****No LLM judge needed****: gold answers are short exact strings (magic numbers, lists of words). Scoring is ``gold in response`` (case-insensitive substring match, all gold answers must appear for multi-value tasks).
- ****A and B are derived from the SAME context****: B is a programmatic excerpt of A's text, not a separately-curated source. This isolates the length-pressure effect cleanly.

Three length levels

- level · context tokens · items per task ·
- --- · --- · --- ·
- L1 · ~4096 · 20 ·
- L2 · ~8192 · 20 ·
- L3 · ~16384 · 20 ·

Three tasks (niah-only — deterministic B)

- ``niah_single_1``: single magic number in haystack
- ``niah_multikey_2``: magic number tagged with a specific key among multiple keys
- ``niah_single_3``: another single-needle variant

180 evaluations total (3 levels × 3 tasks × 20 items).

Two models

- ``deepseek/deepseek-v4-pro`` via DeepSeek API (1M context)
- ``ibm-granite/granite-4.1-8b`` via OpenRouter (131k context)

Pre-registered predictions

- level · predicted A_recall · predicted B_recall · reasoning ·
 - ---- · ---- · ---- · ---- ·
 - 4k · both ≥ 0.85 · both ≥ 0.90 · length not yet pressing for either model ·
 - 8k · DS ≥ 0.80 / Granite ≥ 0.60 · both ≥ 0.85 · Granite shows first signs of pressure ·
 - 16k · DS ≥ 0.75 / Granite ≥ 0.40 · both ≥ 0.85 · Granite drops further; DS still strong ·
- **Hypothesis:** $A \rightarrow B$ improvement Δ grows monotonically with length** on Granite. On DeepSeek v4 Pro the effect may be small at these lengths (we're well within its 1M context window).

Falsification

Hypothesis is REFUTED if:

- B's accuracy is $\leq$ A's on either model at the 16k level
- B's accuracy varies significantly with context length (it should be near-constant — B is always the same ~250-token excerpt)
- Granite A at 16k > 0.80 (length pressure doesn't kick in)

What this run does NOT prove

- ****Synthetic**** — needle-in-haystack is not real conversational memory. Different mechanism from LongMemEval; complementary, not redundant.
- Three needle-type tasks only. Other RULER tasks (qa, vt, cwe, fwe) excluded because B is not deterministically extractable from them.
- Two models only. No Sonnet/GPT-4o/Gemini.
- Length stops at 16k because `simonjegou/ruler` (the public version) has only 4k/8k/16k splits. `MaxJeblick/Ruler` would give 32k/65k/131k but is gated.

Cost and time estimate

- 180 items $\times$ 4 calls (2 models $\times$ 2 variants) = 720 API calls
- Roughly \$3-6 total (DeepSeek dominates because A inputs are 4k-16k tokens)
- Roughly 15-25 minutes wall time at parallelism 4 on GitHub Actions Ubuntu runner

Methodological commitments

- All 180 items run, no cherry-picking. Failed items recorded as score 0.
- Same scoring rule (substring match) across both models and both variants.
- B extraction algorithm pre-committed in `ruler\_run.py` — never changed after seeing results.
- Resume-safe: each item written to disk after both models' both variants complete.
- Hashes of fixtures + extraction algorithm logged for replay.

RULER-Ext 32k/64k/131k

Pre-registration: RULER-Extended A/B run (32k / 64k / 131k)

****Committed BEFORE the run.****

What this test does

Extends the validated RULER A/B (simonjegou/ruler, 4k/8k/16k) into the extreme-length regime using ****MaxJeblick/Ruler**** (32k / 64k / 131k splits).

Same A/B design as before:

- ****Variant A****: full ``input`` field + question. ~32k / 64k / 131k input tokens.
- ****Variant B****: deterministically extracted needle window — the sentence containing the gold answer plus up to 2 sentences before and after.

Field schema (MaxJeblick differs from simonjegou):

- ``input`` (string, the haystack — was ``context``)
- ``question`` (string)
- ``outputs`` (list of strings, gold answers — was ``answer``)
- ``length`` (int)
- ``answer_start`` (string, where the needle starts — extra info)

Three lengths

· level · context tokens · items per task ·

· --- · --- · --- ·

· L1 · 32768 · 20 ·

· L2 · 65536 · 20 ·

· L3 · 131072 · 20 ·

Three tasks (niah-only — deterministic B-extraction)

- ``niah_single_1``
- ``niah_multikey_2``
- ``niah_single_3``

180 evaluations total (3 levels × 3 tasks × 20 items × 2 models × 2 variants

= ****720 API calls****).

Two models

- ``deepseek/deepseek-v4-pro`` via DeepSeek API (1M context — well within range for all 3 levels)
- ``ibm-granite/granite-4.1-8b`` via OpenRouter (131k context — ****at its hard limit at L3****)

Granite is expected to fail (or sharply degrade) at L3 because the full context + system prompt + question + `answer_prefix` exceeds its window.

This is intentional: it tests whether B (~250 tokens) preserves recall when A literally cannot fit.

Pre-registered predictions

· level · DS A · DS B · Granite A · Granite B ·

· --- · --- · --- · --- · --- ·

· 32k · ≥ 0.70 · ≥ 0.85 · ≥ 0.30 · ≥ 0.85 ·

· 64k · ≥ 0.55 · ≥ 0.85 · ≥ 0.10 · ≥ 0.85 ·

· 131k · ≥ 0.40 · ≥ 0.85 · ≈ 0 (over context) · ≥ 0.85 ·

Pre-registered hypotheses

1. **Δ Granite** grows monotonically with length across 32k/64k/131k and exceeds +0.50 at 131k.
2. **Δ DS** becomes meaningfully positive at 131k ($\geq +0.15$). At 16k in the prior run it was +0.083; at 131k length pressure should be strong even for DS v4 Pro.
3. **B** accuracy is near-constant across all 3 lengths — band width ≤ 0.10 on either model. B always sees the same ~250-token excerpt.

Falsification

Hypotheses are REFUTED if any of:

- **B** varies by more than 0.10 across the 3 lengths on either model (would mean B is not actually a length-independent excerpt).
- **DS A** at 131k > 0.75 (would mean DS feels no pressure at 131k — hypothesis 2 fails).
- **Δ Granite** at 131k $< +0.30$ (would mean B doesn't rescue Granite even at extreme over-limit — hypothesis 1 fails badly).
- **Δ Granite** is NOT monotonically non-decreasing across the three lengths (would mean the pressure-relief curve is non-monotonic).

What this run does NOT prove

- Synthetic needle-in-haystack $\neq$ real conversational memory. Complementary to LongMemEval, not redundant.
- Three niah tasks only.
- Two models only.
- Granite at 131k is at/over its hard limit — we cannot disentangle "model ran out of context window" from "model couldn't attend across that many tokens". A score of 0 at 131k could be either. This is fine for the practical claim ("B saves you when A fails") but not for the mechanistic claim.

Cost and time estimate

- 180 items $\times$ 4 calls (2 models $\times$ 2 variants) = 720 API calls
- DS A inputs: $(32k + 65k + 131k) \times 60 = 13.6M$ tokens $\times$ $\$0.43/M = \sim \5.85
- DS B inputs: $\sim 250 \times 180 = 45k \times \$0.43/M \approx \$0.02$
- Granite A inputs: same as DS A but cheaper: $\sim \$0.68$
- Granite B: negligible
- Output tokens (max 128 each): tiny
- **Expected total: \$6–9. Generous upper bound: \$12.**
- Wall time at parallelism 4 on GitHub Actions: $\sim 30\text{--}45$ min.

Methodological commitments

- All 180 items run, no cherry-picking. Failed items recorded as score 0 (including Granite-at-131k context-overflow failures).
- Same scoring rule (substring match) across both models and both variants as in the original RULER run.
- B extraction algorithm is the same as the original — pre-committed in

the runner, never changed after seeing results.

- Resume-safe: each item written to disk after both models' both variants complete.
- Per-item records stored in workflow artifact `ruler-ext-results`.

Zusammenfassung & Caveats

Was die vier Runs zusammen zeigen: Die A/B-Differenz zwischen Volltext-Kontext und deterministischem Excerpt/State (B) ist (a) reproduzierbar positiv über zwei sehr unterschiedliche Benchmarks (synthetisch und real), (b) skaliert monoton mit Kontextlänge auf beiden Modellen, und (c) erreicht dramatische Werte (+0.867 für Granite bei 131k), wenn die Volltext-Strategie physikalisch unmöglich wird. Der Jury-Pilot bestätigt zusätzlich, dass der verwendete Einzel-Judge robust genug für diese Schlussfolgerungen ist.

Was diese Belege NICHT beweisen:

- RULER ist synthetisch (Needle-in-Haystack) — komplementär zu LongMemEval, nicht gleichwertig zu echter Konversations-Memory.
- Nur 3 niah-Tasks getestet. RULER hat 13 Tasktypen; non-niah (qa, vt, cwe, fwe) brauchen task-spezifische B-Extraktion.
- Nur 2 Modelle (DeepSeek v4 Pro, Granite 4.1 8B). Sonnet/GPT-4o/Gemini auf RULER ungetestet.
- Granite-A bei 131k mischt 'Modell kann nicht reinpassen' mit 'Modell kann nicht über die Länge attendieren' — für den praktischen Claim ('B rettet wenn A scheitert') ausreichend, für den mechanistischen Claim ('Attention degradiert mit Länge') nicht.
- Die LongMemEval-strikte-Hypothese ($\Delta_{DS} \geq 0.15$) wurde widerlegt (tatsächlich +0.104). Honest disclosure: DS profitiert weniger stark als gehofft.
- RULER-Ext-H3 (B konstant innerhalb 0.10) wurde strikt widerlegt auf Granite (Bandbreite 0.133). Variation ist non-monoton und vermutlich Sampling-Rauschen, aber der vorab-committete Falsifizierer ist getriggert.

Replay-Fähigkeit

Alle Pre-Registrierungen, Runner-Scripts, Pro-Item-Resultate und Reports sind im Repository `hst re/DESi`, Branch `desi-ruler-bench`, festgeschrieben. Die Pro-Item-RULER-Resultate liegen in den GitHub-Actions-Artifacts (`ruler-results`, `ruler-ext-results`, 30 Tage Retention). B-Extraktion und Scoring sind deterministisch. Modell-Outputs sind nicht seedable auf diesen APIs — Re-Runs liefern sehr ähnliche, aber nicht bit-identische Resultate.